Proof. Note that $1-e$ is also an idempotent in A. Thus the statement immediately follows from the fact that $\mathrm{Hom}_A(A,M) \cong M$ and the decomposition $A = Ae \oplus A(1-e)$. $\square$

Now we are ready to prove Theorem 5.12.2. Let $\lambda \geq \mu$. Then by Lemmas 5.13.3 and 5.13.4

$$\mathrm{Hom}_{S_n}(V_\lambda, V_\mu) = \mathrm{Hom}_{S_n}(\mathbb{C}[S_n]c_\lambda, \mathbb{C}[S_n]c_\mu) = c_\lambda \mathbb{C}[S_n]c_\mu.$$

The latter space is zero for $\lambda > \mu$ by Lemma 5.13.2 and 1-dimensional if $\lambda = \mu$ by Lemmas 5.13.1 and 5.13.3. Therefore, V_λ are irreducible, and V_λ is not isomorphic to V_μ if $\lambda \neq \mu$. Since the number of partitions equals the number of conjugacy classes in S_n, the representations V_λ exhaust all the irreducible representations of S_n. The theorem is proved.

5.14. Induced representations for S_n

Denote by U_λ the representation $\mathrm{Ind}_{P_\lambda}^{S_n}\mathbb{C}$. It is easy to see that U_λ can be alternatively defined as $U_\lambda = \mathbb{C}[S_n]a_\lambda$.

Proposition 5.14.1. *We have* $\mathrm{Hom}(U_\lambda, V_\mu) = 0$ *for* $\mu < \lambda$ *and* $\dim \mathrm{Hom}(U_\lambda, V_\lambda) = 1$. *Thus,* $U_\lambda = \bigoplus_{\mu \geq \lambda} K_{\mu\lambda} V_\mu$, *where* $K_{\mu\lambda}$ *are nonnegative integers and* $K_{\lambda\lambda} = 1$.

Definition 5.14.2. The integers $K_{\mu\lambda}$ are called the **Kostka numbers**.

Proof. By Lemmas 5.13.3 and 5.13.4,

$$\mathrm{Hom}(U_\lambda, V_\mu) = \mathrm{Hom}(\mathbb{C}[S_n]a_\lambda, \mathbb{C}[S_n]a_\mu b_\mu) = a_\lambda \mathbb{C}[S_n]a_\mu b_\mu,$$

and the result follows from Lemmas 5.13.1 and 5.13.2. $\square$

Now let us compute the character of U_λ. Let $C_{\mathbf{i}}$ be the conjugacy class in S_n having i_m cycles of length m for all $m \geq 1$ (here $\mathbf{i}$ is a shorthand notation for $(i_1, \dots, i_m, \dots)$). Also let $x_1, \dots, x_N$ be variables, and let

$$H_m(x) = \sum_j x_j^m$$

be the power sum polynomials.